

Cell Structure and Organisation Main Note

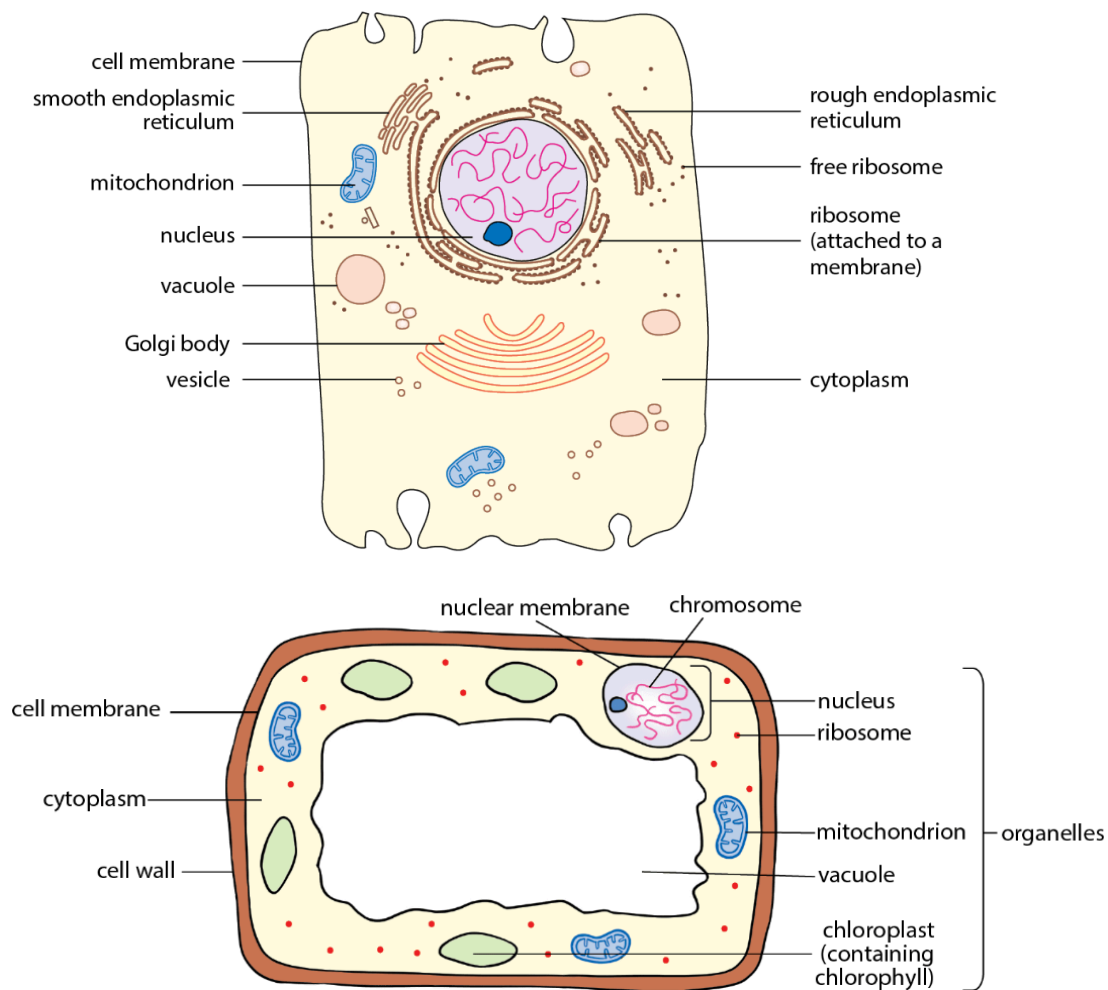


Figure 1.5 A generalised plant cell

What Are Cells:

- All living things made up of billions of tiny cells

Cells: Building blocks of life $\Rightarrow$ simplest structural and functional units of life

What do cells do:

- take in raw materials
- processes these raw materials to make new molecules
- either used by the cell itself or transported to other parts of the body

Cells have diff structures that perform different roles in the cell

- division of labor $\Rightarrow$ increases efficiency within the system
- ensures that cell can survive and perform its role within the body

What Does a Cell Consist of?

Definition: Cell is a unit of life. It consists of a mass of living matter called protoplasm

[#BiologyDefinitions](#)

[Protoplasm](#)

[Cell Membrane](#)

[Cell Wall](#)

[Cytoplasm](#)

[Nucleus](#)

[Rough Endoplasmic Reticulum](#)

[Smooth Endoplasmic Reticulum](#)

[Mitochondria](#)

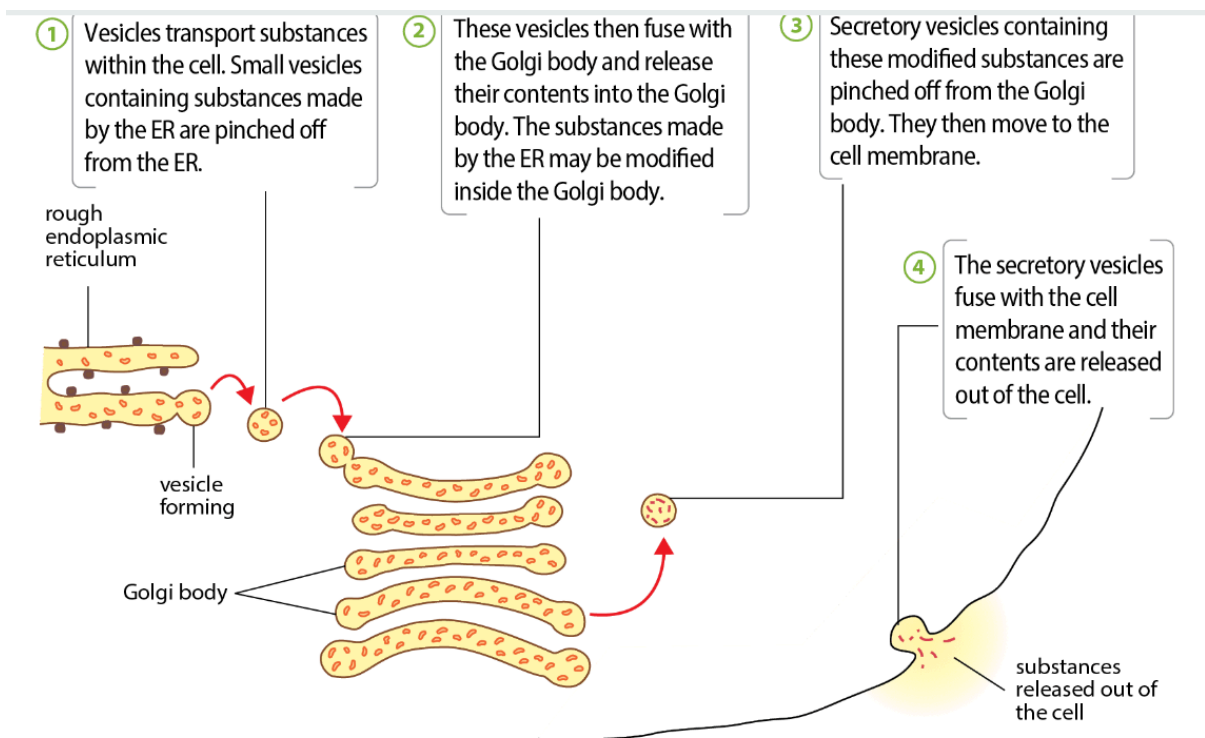
[Ribosomes](#)

[Chloroplasts](#)

[Vacuole](#)

Movement of Substances from ER to out of the cell

1. vesicles transport substances within the cell
⇒ small vesicles ⇒ containing substances made by the ER ⇒ pinched off from the ER
2. These vesicles ⇒ fuse with the Golgi Body
⇒ substances made by the ER might be chemically modified in the Golgi body
3. Secretory Vesicles ⇒ containing chemically modified substances ⇒ pinched off from the Golgi Body
⇒ move to the cell membrane
4. Secretory Vesicles ⇒ fuse with cell membrane
⇒ contents are released outside of the cell



How are cells adapted to their function

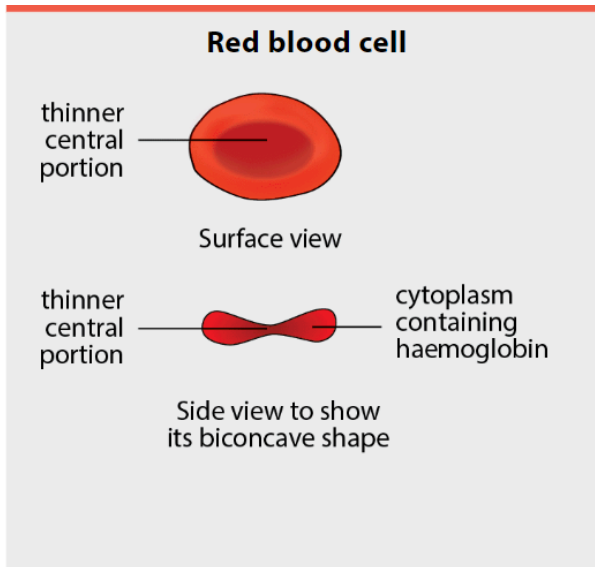
Definition: differentiation is the process by which a cell becomes specialized for a specific function

#BiologyDefinitions

1. RBC

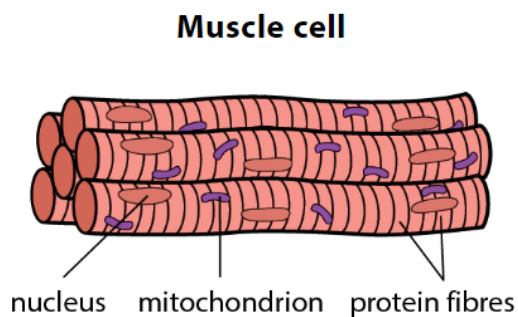
- Contains hemoglobin - binds reversibly to oxygen - transports it around the body
- Circular biconcave shape - increase SA to V ratio - oxygen can diffuse in and out of the cell at a higher rate
- Absence of nucleus - allows for more packing of hemoglobin - more transport of oxygen as more oxygen can fit in the cell. ++

- Flexible - can squeeze through capillaries easily



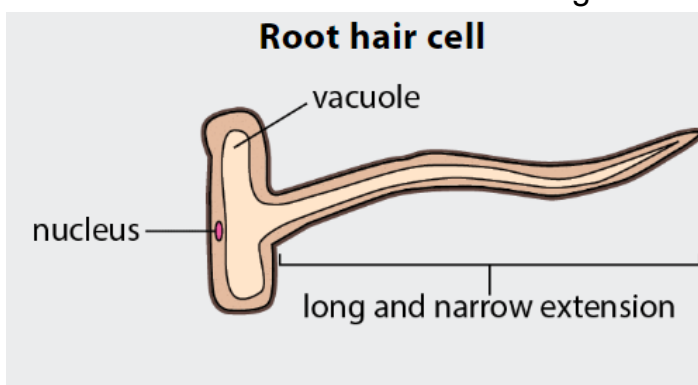
2. Muscle Cell

- Elongated and cylindrical shape, contains many nuclei, contains many mitochondria
- Contains many mitochondria - provide energy for contraction of the muscle cell.
- Has contractile protein fibers - contract and relax to bring about movement
- Many nuclei - allow for cell division.



3. Root Hair Cell

- Long and narrow protrusion/extension → increase SA to V ratio of cell to soil → absorb water and mineral salts at a higher rate.



Note: Mitochondria, ER, Ribosomes, Golgi Body ⇒ only visible under electron microscope